

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```

SQLite Playground // new.db
open file open url settings i

SELECT
  state,
  COUNT(customer_id) AS total_customers,
  ROUND(
    COUNT(customer_id) * 100.0 /
    SUM(COUNT(customer_id)) OVER (),
    2
  ) || '%' AS percentage_distribution
FROM customer_t
GROUP BY state
ORDER BY total_customers DESC
limit 10

run share tables ask AI
sqlite 3.45.0

```

10 rows, took 2 ms

Output:

10 rows, took 2 ms

state	total_customers	percentage_distribution
California	97	9.76%
Texas	97	9.76%
Florida	86	8.65%
New York	69	6.94%
District of Columbia	35	3.52%
Colorado	33	3.32%
Ohio	33	3.32%
Alabama	29	2.92%
Washington	28	2.82%
Arizona	26	2.62%

Observations:

- ~994 customers have placed orders across multiple states
- California and Texas are the largest markets, each contributing ~10% of total customers
- Florida (8.65%) and New York (6.94%) are the next key contributors
- The top 4 states contribute ~35% of the total customer base
- Remaining states individually contribute below 4%, indicating a long-tail distribution

Insights:

- Customer demand is **concentrated in major states**, making them critical to revenue stability
- Any **negative after-sales experience in top states** can significantly impact overall sales trends
- Mid- and low-performing states present **growth opportunities** through targeted expansion and service improvement
- Diversifying customer acquisition across states can **reduce dependency on a few key markets**

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SQLite Playground // new.db

SELECT
    vehicle_maker AS top_vehicle_makers,
    COUNT(customer_id) AS total_customers
FROM product_T JOIN customer_t
GROUP BY 1
ORDER BY 2 DESC
LIMIT 5;
```

run share tables ask AI sqlite 3.45.0

5 rows, took 378 ms

Output:

5 rows, took 378 ms

top_vehicle_makers	total_customers
Chevrolet	82502
Ford	62622
Toyota	51688
Pontiac	49700
Dodge	49700

Observations:

- Chevrolet has the highest customer base (~82.5K), leading by a significant margin
- Ford (~62.6K) and Toyota (~51.7K) are the next major contributors
- Pontiac and Dodge have identical customer counts (~49.7K each)
- The top 3 vehicle makers account for the most of customers, indicating strong brand concentration
- Customer distribution drops noticeably after the top-ranked brand

Insights:

- Chevrolet appears to have strong brand penetration and customer preference
- Ford and Toyota act as primary competitive challengers in the market
- Pontiac and Dodge likely target similar customer segments, suggesting potential market overlap
- Heavy reliance on top vehicle makers could create market dependency risks
- Lower-ranked makers present potential growth opportunities through targeted marketing and product differentiation

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```

SQLite Playground // new.db
open file open url settings

SELECT *
FROM
(
    SELECT
        state,
        vehicle_maker,
        COUNT(customer_id) AS total_customers,
        RANK() OVER (PARTITION BY state ORDER BY COUNT(customer_id) DESC) AS ranking
    FROM product_t
    JOIN order_t USING(product_id)
    JOIN customer_t USING(customer_id)
    GROUP BY 1, 2
) AS preferred_vehicle
WHERE ranking = 1
ORDER BY 3 DESC
limit 15

run share tables ask AI
sqlite 3.45.0

15 rows, took 6 ms

```

Output:

15 rows, took 6 ms

state	vehicle_maker	total_customers	ranking
Texas	Chevrolet	9	1
Florida	Toyota	7	1
California	Nissan	6	1
California	Ford	6	1
California	Dodge	6	1
California	Chevrolet	6	1
California	Audi	6	1
Ohio	Chevrolet	6	1
Alabama	Dodge	5	1
Colorado	Chevrolet	5	1
Maryland	Ford	5	1
New York	Toyota	5	1
New York	Pontiac	5	1
Virginia	Ford	5	1
Washington	Chevrolet	5	1

Note: The actual output contains 143 records; however, it has been restricted to 15 records due to space limitations

Observations:

- Chevrolet appears as the top-ranked vehicle maker across multiple states, indicating strong geographic presence
- Toyota and Ford consistently emerge as major competing brands across several high-volume states
- Large states such as California and New York show diversified customer preference with multiple brands sharing top rank
- Several smaller states show fragmented demand with multiple vehicle makers tied at low customer counts



Insights:

- Chevrolet demonstrates strong nationwide brand penetration and customer acceptance
- Toyota and Ford act as primary competitive challengers in key markets
- Highly competitive states indicate lower brand loyalty and require differentiation-focused marketing strategies
- Fragmented markets present potential growth opportunities through targeted regional marketing and expansion strategies

Question 4: Find the overall average rating given by the customers.

What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
SQLite Playground // new.db
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WITH rating AS (
  SELECT
    customer_feedback,
    quarter_number,
    CASE
      WHEN LOWER(TRIM(customer_feedback)) = 'very bad' THEN 1
      WHEN LOWER(TRIM(customer_feedback)) = 'bad' THEN 2
      WHEN LOWER(TRIM(customer_feedback)) = 'okay' THEN 3
      WHEN LOWER(TRIM(customer_feedback)) = 'good' THEN 4
      WHEN LOWER(TRIM(customer_feedback)) = 'very good' THEN 5
      ELSE 0
    END AS total_rating
  FROM order_t
)
SELECT
  quarter_number,
  ROUND(AVG(total_rating), 2) AS average_rating
FROM rating
GROUP BY quarter_number
ORDER BY quarter_number ASC;

run share tables ask AI
sqlite 3.45.0

4 rows, took 4 ms
```

Output:

4 rows, took 4 ms

quarter_number	average_rating
1	3.55
2	3.35
3	2.96
4	2.4

Observations:

- Average customer rating declines steadily across quarters, dropping from **3.55 in Q1 to 2.40 in Q4**
- Q1 records the highest customer satisfaction level, indicating relatively strong service or product
- The largest decline occurs between **Q3 (2.96) and Q4 (2.40)**, suggesting a significant drop in customer sentiment in the final quarter
- From Q2 onwards, average ratings consistently fall below 3.5, indicating moderate to declining satisfaction levels over time

Insights:

- The continuous downward trend signals potential deterioration in service quality, product performance, or customer experience across the year
- The sharp drop in Q4 suggests operational, seasonal, or delivery-related challenges that may require immediate investigation
- Maintaining Q1 service standards across subsequent quarters could help stabilize customer satisfaction levels
- Proactive feedback analysis and corrective action planning are required to prevent further decline and improve customer retention

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

SQLite Playground // new.db

```
SELECT
  quarter_number,
  SUM(CASE WHEN LOWER(TRIM(customer_feedback)) = 'very good' THEN 1 ELSE 0 END) AS very_good,
  SUM(CASE WHEN LOWER(TRIM(customer_feedback)) = 'good' THEN 1 ELSE 0 END) AS good,
  SUM(CASE WHEN LOWER(TRIM(customer_feedback)) = 'okay' THEN 1 ELSE 0 END) AS okay,
  SUM(CASE WHEN LOWER(TRIM(customer_feedback)) = 'bad' THEN 1 ELSE 0 END) AS bad,
  SUM(CASE WHEN LOWER(TRIM(customer_feedback)) = 'very bad' THEN 1 ELSE 0 END) AS very_bad
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

run share tables ask AI

sqlite 3.45.0

4 rows, took 10 ms

Output:

4 rows, took 10 ms

quarter_number	very_good	good	okay	bad	very_bad
1	93	89	59	35	34
2	75	58	53	37	39
3	38	48	50	52	41
4	20	20	40	58	61

Observations:

- Customer satisfaction declines consistently across quarters, with average ratings dropping from 3.55 in Q1 to 2.40 in Q4
- Q1 shows the strongest customer sentiment, with the highest share of positive feedback (Very Good + Good)
- Negative feedback (Bad + Very Bad) increases steadily over time, overtaking positive feedback by Q3
- The steepest deterioration occurs between Q3 and Q4, driven by a sharp rise in Very Bad ratings

Insights:

- The sustained downward trend indicates progressive deterioration in customer experience across the year
- The sharp Q4 decline suggests operational, seasonal, or delivery-related issues that require urgent review
- Restoring and maintaining Q1 service standards could help reverse declining customer sentiment
- Early detection of negative feedback trends and timely corrective actions are critical to prevent further customer churn

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

SQLite Playground // new.db

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```
SELECT
    quarter_number,
    COUNT(order_id) AS total_orders
FROM order_t
GROUP BY 1
ORDER BY 1;
```

[run](#) [share](#) [tables](#) [ask AI](#)

sqlite 3.45.0

4 rows, took 13 ms

Output:

4 rows, took 13 ms

quarter_number	total_orders
1	310
2	262
3	229
4	199

Observations:

- Total orders show a consistent quarter-on-quarter decline, dropping from **310 in Q1 to 199 in Q4**
- **Q1 records the highest order volume**, indicating strong demand at the beginning of the year
- Orders decrease steadily each quarter, with no recovery or stabilization phase observed
- The **largest absolute drop occurs between Q1 and Q2**, followed by continued softening through Q4

Insights:

- The sustained downward trend suggests **weakening customer demand or engagement over time**
- Absence of any rebound indicates that underlying issues were **not corrected early in the year**
- The sharp early decline highlights the need for **proactive monitoring and timely interventions**
- Re-examining Q1 strategies (pricing, promotions, service levels) could help **arrest and reverse the declining order trend**

Question 7: Calculate the net revenue generated by the company.

What is the quarter-over-quarter % change in net revenue?

Solution Query:

SQLite Playground // new.db

```
WITH QoQ AS
(
    SELECT quarter_number,
    ROUND(SUM(quantity * (vehicle_price - ((discount/100)*vehicle_price))), 0) AS revenue
    FROM order_t
    GROUP BY quarter_number)
SELECT quarter_number, revenue,
ROUND(LAG(revenue) OVER(ORDER BY quarter_number), 2) AS previous_revenue,
ROUND((revenue - LAG(revenue) OVER(ORDER BY quarter_number))/LAG(revenue) OVER(ORDER BY quarter_number), 2) AS qoq_perc_change
FROM QoQ;
```

run share tables ask AI

sqlite 3.45.0

4 rows, took 13 ms

Output:

4 rows, took 13 ms

quarter_number	revenue	previous_revenue	qoq_perc_change
1	39421580		
2	32715830	39421580	-0.17
3	29229896	32715830	-0.11
4	23346780	29229896	-0.2

Observations:

- Revenue shows a consistent quarter-on-quarter decline, falling from **39.4M in Q1 to 23.3M in Q4**
- Q1 generates the highest revenue**, setting a strong baseline at the start of the year
- Revenue drops by **17% in Q2**, followed by a **11% decline in Q3**
- The **steepest decline occurs in Q4**, with a **20% QoQ drop**, indicating accelerated revenue erosion

Insights:

- The sustained negative QoQ growth highlights **persistent revenue pressure throughout the year**
- The sharper decline in Q4 suggests **compounding effects of lower demand, pricing pressure, or higher discounting**
- Early revenue contraction in Q2 indicates that **corrective measures were either delayed or ineffective**
- Stabilizing revenue requires **addressing both volume decline and discount strategy**, particularly before entering later quarters

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SQLite Playground // new.db
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SELECT
    quarter_number,
    ROUND(SUM(quantity*vehicle_price), 0) AS revenue,
    COUNT(order_id) AS total_order
FROM order_t
GROUP BY 1
ORDER BY 1;

run share tables ask AI sqlite 3.45.0

4 rows, took 12 ms
```

Output:

4 rows, took 12 ms

quarter_number	revenue	total_order
1	39637631	310
2	32913738	262
3	29435427	229
4	23496008	199

Observations:

- Both **revenue and order volume decline consistently** across quarters, indicating a parallel downward trend
- Q1 records the highest performance** with ~39.6M revenue and 310 orders, serving as the annual peak
- Revenue decreases in line with falling order counts, suggesting **volume-driven revenue contraction**
- Q4 shows the lowest revenue (~23.5M) and order volume (199)**, marking the weakest quarter of the year

Insights:

- The strong alignment between declining orders and revenue indicates that **volume loss is the primary driver of revenue erosion**, rather than pricing alone
- The absence of any quarter-level stabilization suggests **structural demand or engagement issues** rather than a one-time disruption
- Q4's weak performance reflects **compounded effects of sustained demand decline**, making recovery more difficult in later periods
- Improving revenue will require **reversing order volume trends first**, supported by targeted demand-generation and retention strategies

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

SQLite Playground // new.db

```
SELECT
    credit_card_type,
    ROUND(AVG(discount) * 100, 2) || '%' AS average_discount
FROM order_t t1
INNER JOIN customer_t t2
    ON t1.customer_id = t2.customer_id
GROUP BY credit_card_type
ORDER BY AVG(discount) DESC;
```

run share tables ask AI sqlite 3.45.0

16 rows, took 3 ms

Output:

16 rows, took 3 ms

credit_card_type	average_discount
laser	64.38%
mastercard	62.95%
maestro	62.42%
visa-electron	62.35%
china-unionpay	62.22%
instapayment	62.06%
americanexpress	61.63%
diners-club-us-ca	61.46%
diners-club-carte-blanche	61.45%
switch	61.02%
bankcard	60.95%
jcb	60.74%
visa	60.08%
diners-club-enroute	59.98%
solo	58.5%
diners-club-international	58.4%

Observations:

- Average discounts remain **consistently high across all credit card types** (≈58%–64%)
- **Laser, Mastercard, and Maestro** receive the highest average discounts
- **Minimal variation across card types** indicates little differentiation in discounting

Insights:

- The uniform discount pattern suggests a **non-targeted discount strategy**
- High discounts across all cards may be **pressuring margins without clear payoff**
- **Card-specific discount optimization** could improve profitability while maintaining demand

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

 SQLite Playground // new.db

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```
SELECT
    quarter_number,
    ROUND(AVG(julianday(ship_date) - julianday(order_date)), 0)
        AS average_shipping_time
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

[run](#) [share](#) [tables](#) [ask AI](#)

sqlite 3.45.0

4 rows, took 9 ms

Output:

4 rows, took 9 ms

quarter_number	average_shipping_time
1	57
2	71
3	118
4	174

Observations:

- Average shipping time **increases sharply quarter over quarter**, rising from **57 days in Q1 to 174 days in Q4**
- Each quarter shows a **significant deterioration**, with no signs of stabilization
- **Q3 to Q4 records the steepest increase**, indicating severe late-year delays

Insights:

- The consistent rise in shipping time suggests **progressive operational or logistics breakdowns**
- Prolonged shipping delays likely contribute to **declining customer satisfaction and order volume**
- Immediate intervention is required to **identify bottlenecks and restore shipping SLAs**, especially before peak periods

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
~125M	1000	994	3.13
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
~23.5M	199	98	44.1%

Executive Summary

- Business performance declined steadily across the year, with total orders falling from 310 in Q1 to 199 in Q4 and revenue dropping from ~39.6M to ~23.5M in the last quarter
- Customer experience deteriorated significantly, reflected in declining average ratings (3.55 → 2.40), low good-feedback share (44.1%), and sharply increasing shipping delays (57 days → 174 days)
- Shipping delays emerged as a critical driver, strongly correlating with reduced customer satisfaction, lower repeat demand, and revenue erosion
- Q4 represents the weakest quarter across all key metrics, indicating compounded operational and customer-experience issues rather than isolated disruptions
- Immediate focus on fulfilment efficiency, customer feedback management, and product mix optimization is required to stabilize demand and prevent further revenue loss

Business Recommendations

- Prioritize improvements in **shipping and fulfilment**, as rising delivery time (98 days) strongly correlates with declining ratings (average 3.13) and feedback quality
- Address **customer experience gaps** by analyzing feedback trends and acting early on negative sentiment signals (only 44.1% good feedback)
- Investigate the **sharp Q4 order decline (199 orders)** to identify demand loss driven by service delays or post-purchase experience
- Break down **last-quarter revenue decline (23.5M)** to distinguish volume-driven losses from pricing or discount impact
- Optimize the **product mix** by strengthening supply for top-selling items and reassessing consistently low-performing products to prevent revenue leakage